EXPERIMENT 6

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

Time Series Analysis of Website Traffic

Aim: Identify trends and anomalies in daily website traffic.

Python Program

```
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv('website_traffic.csv',parse_dates=['Date']).set_index('Date')
df['Page_Views']=df['Page_Views'].interpolate()
df['Page_Views'].plot(); plt.show()
df['MA']=df['Page_Views'].rolling(7).mean()
df[['Page_Views','MA']].plot(); plt.show()
z=(df.Page_Views-df.Page_Views.mean())/df.Page_Views.std()
print(df[z.abs()>3])
```

Result

Traffic trends and moving averages are plotted. Values with absolute Z-score above 3 are flagged as possible anomalies.

Note: For CSV programs, keep the named CSV file in the same folder and check that column names match.